Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/16/2024
PI Name:	Ramasubramanian, Anand Ph.D.		
Project Title:	Thrombosis in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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PI Organization Type:	UNIVERSITY	**Phone:**	408-924-3922
Organization Name:	San Jose State University Research Foundation		
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City:	San Jose	**State:**	CA
Zip Code:	95192-1000	**Congressional District:**	19
Comments:			
Project Type:		**Solicitation / Funding Source:**	
Start Date:	12/01/2020	**End Date:**	07/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 07/31/2024 per F. Hernandez/ARC (Ed., 6/18/24) NOTE: End date changed to 11/30/2023 per NSSC information (Ed., 11/2/22) NOTE: End date changed to 11/30/2022 per NSSC information; original end date was 11/30/2021 (Ed. 10/26/21)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Lee, Sang-Joon Ph.D. (San Jose State University Research Foundation) Lee, Wendy Ph.D. (San Jose State University Research Foundation)		
Grant/Contract No.:	80NSSC21K0272		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Dramatic changes in cardiovascular physiology during spaceflight are well-documented: redistribution of fluids from lower to upper part of the body, a remarkable decrease in blood volume, and an increase in stroke volume and cardiac output. The headward fluid shift due to weightlessness results in sustained intracranial pressure. In January 2020, the New England Journal of Medicine (NEJM) report detailed the first ever occlusive thrombotic event in the internal jugular vein (IJV) of an astronaut in the International Space Station, and a 2019 JAMA (Journal of the American Medical Association) study reported that structural changes in the internal jugular vein of astronauts, and fluid congestion due to headward fluid shift as the newly discovered risk factors of stroke during space travel. These studies reveal gaps in our understanding of circulatory and hemostatic physiology during space travel. Our objective of this work is to understand the effect of microgravity and headward fluid shift on the risk of venous thrombosis. We will test the hypothesis that microgravity modulates the risk of cerebral venous thrombosis through effects on endothelial activation and local transport dynamics. Using a scale-down microfluidic endothelialized model of the vasculature subjected to microgravity in a random positioning machine (RPM), we will test the hypothesis through the following specific aims: (i) evaluate the combined effect of microgravity and venous flow on endothelial transcriptome and miRNA-ome; and (ii) evaluate the effect of microgravity on venous thrombosis. We expect that our investigative team with expertise in vascular mechanobiology, microfluidics, and system biology will provide insights into poorly understood aspects of microgravity on cardiovascular physiology, which have been mentioned as an area of Space Biology research emphases, and also as high priority in the midterm assessment of the Decadal report. Further, in our opinion, the recent incidents raise the risk of thrombosis as a health concern in astronauts. We believe that our innovative ground analog will generate testable hypothesis, and new, high-value tools that can be translated to full scale spaceflight investigations on hemostasis and thrombosis. References: Auñón-Chancellor SM, Pattarini JM, Moll S, Sargsyan A. Venous Thrombosis during Spaceflight. N Engl J Med. 2020 Jan 2;382(1):89-90. http://doi.org/ ; PMID: 31893522 . Marshall-Goebel K, Laurie SS, Alferova IV, Arbeille P, Auñón-Chancellor SM, Ebert DJ, Lee SMC, Macias BR, Martin DS, Pattarini JM, Ploutz-Snyder R, Ribeiro LC, Tarver WJ, Dulchavsky SA, Hargens AR, Stenger MB. Assessment of Jugular Venous Blood Flow Stasis and Thrombosis During Spaceflight. JAMA Netw Open. 2019 Nov 1;2(11):e1915011. http://doi.org/ ; PMID: 31722025 ; PMCID: PMC6902784 .
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the effects of hydrodynamic and gravitational forces on the growth and function of endothelial cells, which line the blood vessels, will provide deeper insights into both health and disease. Fundamental insights into how forces may help maintain cardiovascular health, and how perturbations may result in thrombosis, will be attained.
Task Progress:	Our objective of this work is to understand the effect of microgravity and headward fluid shift on the risk of venous thrombosis. We will test the hypothesis that microgravity modulates the risk of cerebral venous thrombosis through effects on endothelial activation and local transport dynamics. We tested the hypothesis using a scale-down microfluidic endothelialized model of the vasculature subjected to microgravity in a random positioning machine (RPM); we evaluated the combined effect of microgravity and flow. We also evaluated the flow effects on thrombus formation in venous valve mimics. 1. Development of an omnidirectional bubble-trap for continuous culture of endothelial cells under flow in microgravity. We optimized and characterized an in-house closed-loop flow system. We had noticed that the closed-loop flow system using commercial pumps and bubble traps were not effective in removing microbubbles which were detrimental to cell culture. Therefore, we developed a novel 3D-printed bubble trap which was much more effective in removing the microbubbles, as confirmed by following output from a pressure sensor. The 3D-printed bubble trap idea and demonstration has been filed for a provisional patent application. We cultured primary human venous endothelial cells under flow conditions in normal and microgravity conditions. Without the bubble trap, cell culture was unsuccessful since the bubbles detached the endothelial monolayer. However, with the bubble trap, continuous culture of the cells in both normal and microgravity over 24 h period was accomplished successfully. We noticed that while shear stress aligned the cells in the direction of flow under normal gravity, no such alignment was seen under microgravity. 2. Characterization of the dynamics of red blood cell (RBC) distribution in venous valve mimics. We developed a microfluidic model to characterize the effects of unique geometry on blood flow patterns and RBC distribution in and around venous valves. We observed that the flow rates contributed to an uneven distribution of RBCs in the vortices and the entrapment of RBCs in the vortices. Further, experimental and computational characterization of flow profiles showed that the lateral movement of RBC towards the vessel walls is dependent on a threshold shear rate below which RBCs do not enter the pocket. A high priority in the midterm assessment of the Decadal report is to use a systems biology approach to understand the underlying mechanisms of microgravity effects on cardiovascular physiology. [Ed. Note: For more information on the Decadal Surveys, visit: https:// .] Recent incidents raise the risk of thrombosis as a health concern in astronauts. Studies, such as this one using ground-based analogs, will generate testable hypotheses and new, high-value tools that can be translated to full-scale spaceflight investigations on the risk of thrombosis.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Articles in Peer-reviewed Journals	Vegunta B, Lui C, Kim M, Tran J, Papez M, Ramasubramanian AK, Lee SJ. "Orientation-independent bubble trap with internal partition for robust operation of microfluidic systems." Lab Chip. 2025 Jul 8;25(14):3423-9. https://doi.org/10.1039/d4lc01060d ; PubMed PMID: 40548853 , Jul-2025
Patents	18/634,723. US patent application filed April 2024. Apr-2024 Lee SJJ, Vegunta B, Ramasubramanian AK, Kim M, Tran J. "Bubble Trap with Fluid Guide."